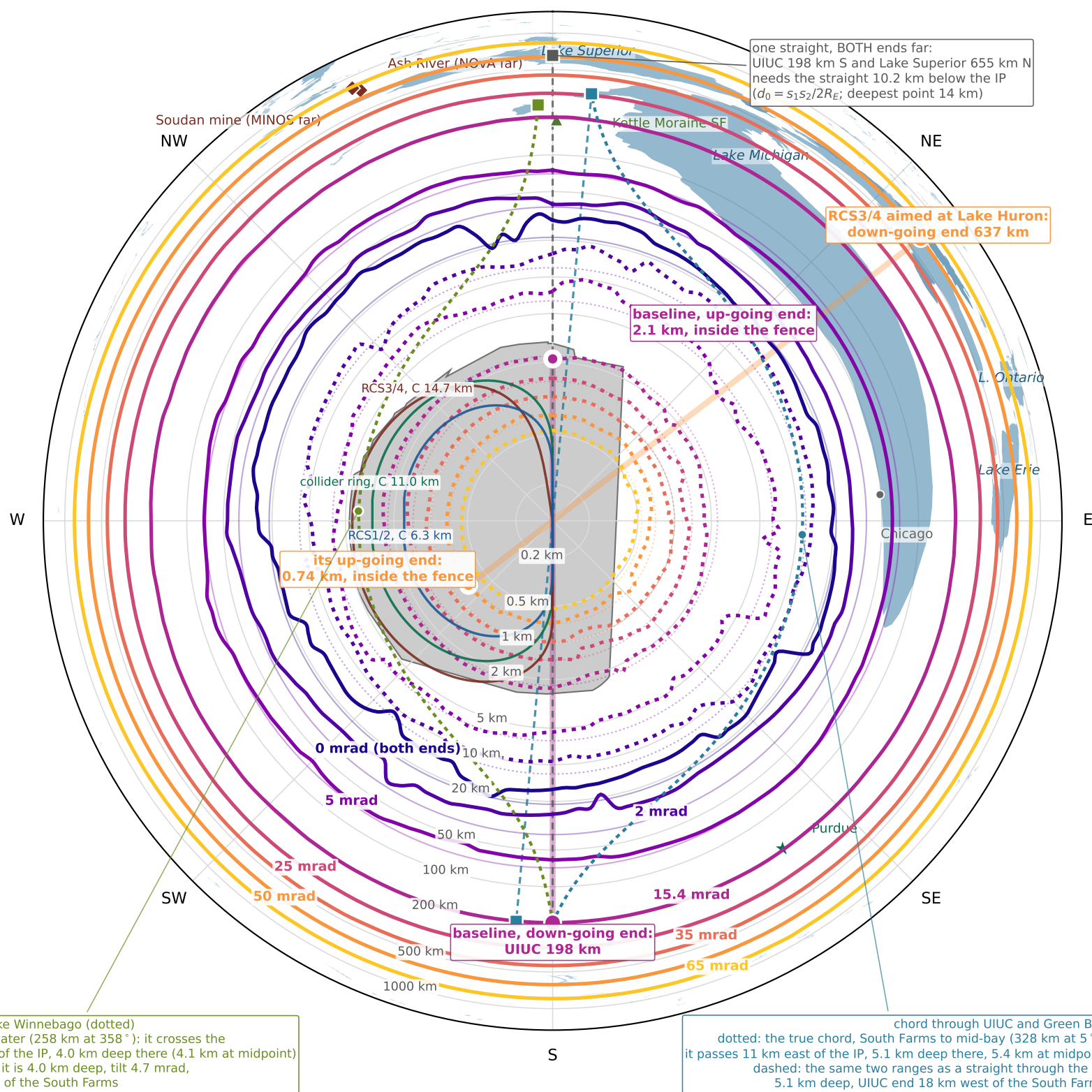


Where the two ends of one straight surface, tilt by tilt
 35 m-deep straight; one colour per tilt – dotted: the end that goes up, solid: the end that goes down; at 0 mrad they are the same curve
 N



- up-going end surfaces here (near emergence)
- down-going end surfaces here (far exit)
- same, smooth sphere
- Fermilab site

- lakes (Natural Earth 10m)
- one straight: its two ends are opposite each other
- the rings in plan (collider, RCS3/4, RCS1/2)
- a chord with both ends far (10 km deep)

- the UIUC—Green Bay chord (true, misses the IP)
- same ranges as a straight through the IP
- the UIUC—Lake Winnebago chord (true, crosses the site)

Bold curves: USGS NED 10 m terrain inside 6 km and GEBCO 2020 beyond, along 144 bearings (0.1–1000 km), Catmull-Rom smoothed; exits over the Great Lakes taken at the water surface. Faint circles: smooth sphere,
 $s_{near} = R_E(-\theta + \sqrt{\theta^2 + 2d_0/R_E}) \approx d_0/\theta$ and $s_{far} = R_E(\theta + \sqrt{\theta^2 + 2d_0/R_E}) \approx 2R_E\theta$. Log range; rings in km. The fence is 0.9–4.9 km out, so the dotted curves show which tilts emerge on site.